

The Exact Normalized History Cocycle

Rectangular Time Evolution, Singular Values, and a Reset-Invariance Counterexample

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July 2026

Abstract

RH-172 realized every positive reset-memory packet canonically in a finite normalized history space. The next question is whether these spaces and packets possess an intrinsic time evolution. We derive that evolution exactly. If $S_{t+1} = AS_t$, $\alpha_t = \|S_t\|_F$, and $r_t = \alpha_t/\alpha_{t+1}$, then the history factors satisfy

$$F_{t+1} = T_t F_t, \quad T_t(y_0, \dots, y_t) = (r_t A y_0, \sqrt{\eta} y_0, \dots, \sqrt{\eta} y_t).$$

Thus the memory recursion is generated by an exact rectangular cocycle $T_t : \mathcal{K}_t \rightarrow \mathcal{K}_{t+1}$.

We compute its adjoint and full singular-value structure:

$$T_t^* T_t = \text{diag}(r_t^2 A^* A + \eta I, \eta I, \dots, \eta I).$$

For $t \geq 1$, T_t is bounded below by $\sqrt{\eta}$ and $\|T_t\| = \sqrt{r_t^2 \|A\|^2 + \eta}$. These formulas are exact and give a well-conditioned finite-time injection even when A is singular.

However, exact factor evolution does not imply that independently selected top-Gram packets form an invariant cocycle bundle. We construct a two-column diagonal example in which the top reset packet rotates by ninety degrees in one step, so the transported and reset history ranges have projection distance one. A 160-case audit verifies the action, adjoint, and singular value formulas to 1.5×10^{-15} and reproduces the exact counterexample. The result closes finite history kinematics, not a transfer-space intertwining or a physical Riesz cloud.

1 From a static realization to time evolution

Let $A \in \mathcal{B}(\mathcal{Y})$, let $S_0 : \mathcal{X} \rightarrow \mathcal{Y}$, and set $S_t = A^t S_0 \neq 0$. With $\alpha_t = \|S_t\|_F$ and $\hat{S}_t = S_t/\alpha_t$, RH-172 defines

$$F_t x = (\hat{S}_t x, \sqrt{\eta} \hat{S}_{t-1} x, \dots, \eta^{t/2} \hat{S}_0 x) \in \mathcal{K}_t := \mathcal{Y}^{\oplus(t+1)}$$

and proves $M_t = F_t^* F_t$ [1]. Since $\mathcal{K}_t$ changes with t , the update is not an ordinary power of one finite square matrix. It is a nonautonomous rectangular cocycle.

The normalization matters. Although $S_{t+1} = AS_t$, generally $\hat{S}_{t+1} \neq A\hat{S}_t$. The missing scalar is

$$r_t = \frac{\alpha_t}{\alpha_{t+1}}, \quad \hat{S}_{t+1} = r_t A \hat{S}_t. \tag{1}$$

2 The rectangular history cocycle

Define $T_t : \mathcal{K}_t \rightarrow \mathcal{K}_{t+1}$ by

$$T_t(y_0, y_1, \dots, y_t) = (r_t A y_0, \sqrt{\eta} y_0, \sqrt{\eta} y_1, \dots, \sqrt{\eta} y_t). \quad (2)$$

Theorem 2.1 (Exact normalized history evolution). *For every $t \geq 0$,*

$$F_{t+1} = T_t F_t. \quad (3)$$

Consequently

$$M_{t+1} = F_t^* T_t^* T_t F_t. \quad (4)$$

Proof. The first block of $T_t F_t x$ is $r_t A \hat{S}_t x = \hat{S}_{t+1} x$ by (1). Every remaining block is the corresponding block of $F_t x$ multiplied by $\sqrt{\eta}$. This is exactly the ordered definition of $F_{t+1} x$. Taking adjoints gives (4). $\square$

The theorem does more than restate the scalar memory recursion. It identifies the full analysis operator that transports every source vector, including phase and cross-coordinate information discarded by M_t alone.

3 Adjoint, norm, and lower bound

For $z = (z_0, \dots, z_{t+1}) \in \mathcal{K}_{t+1}$, direct calculation gives

$$T_t^* z = (r_t A^* z_0 + \sqrt{\eta} z_1, \sqrt{\eta} z_2, \dots, \sqrt{\eta} z_{t+1}). \quad (5)$$

Theorem 3.1 (Complete singular-value formula). *Relative to $\mathcal{K}_t = \mathcal{Y} \oplus \mathcal{Y}^{\oplus t}$,*

$$T_t^* T_t = \text{diag}(r_t^2 A^* A + \eta I, \eta I, \dots, \eta I). \quad (6)$$

Hence

$$\|T_t\| = \sqrt{r_t^2 \|A\|^2 + \eta}. \quad (7)$$

If $t \geq 1$, then

$$\|T_t y\| \geq \sqrt{\eta} \|y\| \quad \text{for every } y \in \mathcal{K}_t. \quad (8)$$

For $t = 0$, the least singular value is $\sqrt{r_t^2 s_{\min}(A)^2 + \eta}$.

Proof. Using (2),

$$\|T_t y\|^2 = r_t^2 \|A y_0\|^2 + \eta \sum_{j=0}^t \|y_j\|^2.$$

There are no cross terms because the outputs occupy distinct direct-sum coordinates. This proves (6). Its largest eigenvalue is $r_t^2 \|A\|^2 + \eta$. When $t \geq 1$, the later diagonal blocks have eigenvalue η , while the first block is bounded below by ηI . The stated formulas follow. $\square$

Thus every positive forgetting weight regularizes the finite rectangular map: T_t is injective with closed range even when A has a kernel. This finite-time fact should not be confused with compactness of an infinite history completion; RH-175 will show that the latter contains a unilateral shift obstruction.

4 Transported packets and two directional defects

Let P_t be a rank- r reset packet for M_t , and let $J_t : \text{Ran } P_t \rightarrow \mathcal{K}_t$ be its canonical polar realization. Choose orthonormal frames V_t and V_{t+1} for the realized ranges. Define the reduced cocycle

$$B_t = V_{t+1}^* T_t V_t. \quad (9)$$

The two natural residuals are

$$R_t^{\text{pr}} = (I - V_{t+1} V_{t+1}^*) T_t V_t, \quad (10)$$

$$R_t^{\text{ad}} = (I - V_t V_t^*) T_t^* V_{t+1}. \quad (11)$$

The first asks whether the old packet transports into the new packet. The second asks whether the new packet pulls back into the old packet. For a nonnormal or rectangular map these are independent. A small primal defect does not imply a small adjoint defect.

Proposition 4.1 (Exact reduction criterion). *The identities*

$$T_t V_t = V_{t+1} B_t, \quad T_t^* V_{t+1} = V_t B_t^*$$

hold simultaneously if and only if $R_t^{\text{pr}} = R_t^{\text{ad}} = 0$. In that case the two realized packet ranges form a reducing pair for the rectangular cocycle.

Proof. Orthogonal decomposition of $T_t V_t$ into the range of V_{t+1} and its orthogonal complement gives the first equivalence. The coefficient of the range component is $V_{t+1}^* T_t V_t = B_t$. Applying the same argument to the adjoint gives the second equivalence and coefficient B_t^* . $\square$

The Gram recursion alone does not imply either residual vanishes. At time $t + 1$, the reset packet optimizes the new quadratic form M_{t+1} ; it is not defined as the image of the old packet under T_t .

5 An exact ninety-degree reset counterexample

Take $\mathcal{X} = \mathcal{Y} = \mathbb{C}^2$, choose $0 < \varepsilon < 1$, and set

$$S_0 = \begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix}, \quad A = \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon^{-1} \end{pmatrix}.$$

Then

$$S_1 = A S_0 = \begin{pmatrix} \varepsilon & 0 \\ 0 & 1 \end{pmatrix}.$$

For every $0 \leq \eta < 1$, the rank-one top packet of $M_0 = \hat{S}_0^* \hat{S}_0$ is $\mathbb{C} e_1$, while the rank-one top packet of $M_1 = \hat{S}_1^* \hat{S}_1 + \eta M_0$ is $\mathbb{C} e_2$.

Theorem 5.1 (Gram optimality does not imply cocycle invariance). *In the preceding example, the canonical realized range of the old top packet after transport by T_0 is orthogonal to the canonical realized range of the new top packet. Therefore*

$$\|P_{T_0 V_0} - P_{V_1}\| = 1.$$

Proof. All matrices are diagonal. The image of the first source coordinate occupies only first-coordinate output axes in both history blocks, whereas the image of the second source coordinate occupies only second-coordinate axes. The two history columns are orthogonal. Polar normalization changes their lengths but not their ranges. Orthogonal rank-one projections have operator distance one. $\square$

The counterexample is not a statement about the physical frozen models. It shows that no theorem can deduce packet invariance solely from positivity, the memory recursion, and exact factor evolution. Additional dynamical structure is indispensable.

6 Block-time cocycle and an explicit long-step formula

The one-step maps compose without ambiguity even though their domains change. For integers $s < t$, define

$$T_{t:s} = T_{t-1}T_{t-2}\cdots T_s : \mathcal{K}_s \rightarrow \mathcal{K}_t.$$

Proposition 6.1 (Exact block-time evolution). *For every $0 \leq s < t$,*

$$F_t = T_{t:s}F_s. \tag{12}$$

Writing $n = t - s$, $q = \sqrt{\eta}$, and $y = (y_0, y_1, \dots, y_s)$, the first $n + 1$ output blocks of $T_{t:s}y$ are

$$q^j \frac{\alpha_s}{\alpha_{t-j}} A^{n-j} y_0, \quad 0 \leq j \leq n, \tag{13}$$

and the remaining blocks are

$$q^n y_1, q^n y_2, \dots, q^n y_s. \tag{14}$$

Proof. Equation (12) follows by iterating Theorem 2.1. For the explicit blocks, induction shows that the j th newly generated block contains j shifts and $n - j$ first-block updates. The product of normalization ratios telescopes:

$$r_s r_{s+1} \cdots r_{t-j-1} = \frac{\alpha_s}{\alpha_{t-j}}.$$

Every old tail block is shifted exactly n times, giving (14). $\square$

The formula separates two mechanisms: physical propagation by powers of A , and memory aging by powers of q . It also gives a type-correct starting point for lagged or block packets. Such packets need not inherit the severe one-step adjoint defect observed later, because their reduction target can be a whole product $T_{t:s}$ rather than one T_s .

7 From primal residual to subspace distance

Let $W = T_t V_t$ and assume W has full column rank. Its polar frame is $\widehat{W} = W(W^*W)^{-1/2}$. The projection distance from the transported range to the new packet is

$$d_t = \|(I - V_{t+1}V_{t+1}^*)\widehat{W}\|.$$

Proposition 7.1 (Conditioned primal-distance bound). *With R_t^{pr} from (10),*

$$d_t \leq \frac{\|R_t^{\text{pr}}\|}{s_{\min}(W)} = \epsilon_t^{\text{pr}} \kappa(W), \quad (15)$$

where $\epsilon_t^{\text{pr}} = \|R_t^{\text{pr}}\| / \|W\|$ and $\kappa(W) = \|W\| / s_{\min}(W)$.

Proof. Since $\widehat{W} = W(W^*W)^{-1/2}$,

$$(I - V_{t+1}V_{t+1}^*)\widehat{W} = R_t^{\text{pr}}(W^*W)^{-1/2}.$$

The inverse positive factor has norm $s_{\min}(W)^{-1}$. Taking norms gives the first inequality; multiplying and dividing by $\|W\|$ gives the second. $\square$

Thus a small normalized primal residual need not force a small subspace distance unless the transported packet is also well conditioned. Conversely, the adjoint residual is not controlled by (15) at all. This quantitative separation motivates recording all three diagnostics in the physical audit rather than treating one as a proxy for the others.

8 Finite identity audit

The implementation audit uses dimensions 2, 3, 5, 8, input history lengths 1, 2, 4, 7, and ten random trials per configuration, for 160 cases. It compares explicit rectangular matrices with blockwise action and adjoint formulas, and compares numerical singular values with Theorem 3.1.

quantity	maximum residual
matrix action	1.48×10^{-15}
adjoint action	4.45×10^{-16}
largest singular value	1.34×10^{-15}
smallest singular value	5.00×10^{-16}
counterexample projection distance	1.000000000000

These are implementation checks, not physical interval certificates.

9 Route consequence and boundary

RH-172 and RH-173 now provide exact finite history kinematics:

$$M_t = F_t^* F_t, \quad F_{t+1} = T_t F_t, \quad T_t^* T_t \text{ explicit.}$$

The next executable question is empirical but type-correct: on the 130 archived physical reset snapshots, how large are the transported packet distance and the two directional residuals? That is the task of RH-174.

Nothing here maps $\mathcal{K}_t$ into the RH-80 transfer space. The maps T_t are rectangular and time dependent, so they do not themselves possess a standard Fredholm determinant. No all-level autonomous completion, Riesz cloud, Schatten limit, self-adjoint generator, zeta divisor, or Riemann-hypothesis claim is made.

References

- [1] Liang Wang. Canonical polar realization of reset-memory packets, 2026. Prime Dynamics Theory Program, RH-172.